

26 Supersymmetric Field Theories

26.5 Spontaneous Supersymmetry Breaking in the Tree Approximation

We saw in the previous section that supersymmetry is unbroken (at least in the tree approximation) in renormalizable theories of chiral superfields if Eq. (26.4.8) has a solution, that is, if there is a value ϕ_0 of the fields at which the superpotential $f(\phi)$ is stationary:

$$\left. \frac{\partial f(\phi)}{\partial \phi_n} \right|_{\phi=\phi_0} = 0. \quad (26.5.1)$$

There are as many independent variables here as there are equations to satisfy, so we generally expect there to be solutions of Eq. (26.5.1). In order for supersymmetry to be spontaneously broken in these theories, it is necessary to impose restrictions on the form of the superpotential.

To see how a choice of superpotential may allow supersymmetry to be spontaneously broken, we will consider a generalization of a class of models due to O’Raifeartaigh [3]. Suppose that the superpotential is a linear combination of a set Y_i of left-chiral superfields, with coefficients given by functions $f_i(X)$ of a second set of left-chiral superfields X_n :

$$f(X, Y) = \sum_i Y_i f_i(X). \quad (26.5.2)$$

The conditions for supersymmetry to be unbroken by the values x_n and y_i of the scalar components of these superfields are that

$$0 = \frac{\partial f(x, y)}{\partial y_i} = f_i(x), \quad (26.5.3)$$

and

$$0 = \frac{\partial f(x, y)}{\partial x_n} = \sum_i y_i \frac{\partial f_i(x)}{\partial x_n}. \quad (26.5.4)$$

Eq. (26.5.4) can always be solved by taking $y_i = 0$, with no effect on the problem of solving Eq. (26.5.3). On the other hand, if the number of X_n superfields is smaller than the number of Y_i superfields, then Eq. (26.5.3) imposes more conditions on the x_n than there are variables, so without fine-tuning a solution is impossible, and supersymmetry is broken.

It may appear that the initial assumption (26.5.2) itself represents a radical form of fine-tuning, but in fact this form can be imposed on the superpotential by assuming a suitable R -symmetry. As discussed in Section 26.3, in theories with $N = 1$ supersymmetry an R -symmetry is a $U(1)$ symmetry for which the θ superspace coordinate has a non-trivial transformation property. If we assume an R -symmetry for which θ_L carries quantum number $+1$, then the $\mathcal{F}$ -term of any superpotential has a quantum number equal to that of the superpotential itself minus 2, so R invariance requires the superpotential itself to have $R = 2$. We can therefore impose the structure (26.5.2) by requiring R invariance, with the Y_i and X_n superfields given R quantum numbers $+2$ and 0 , respectively.

The scalar fields in this sort of model have a potential

$$V(x, y) = \sum_i |f_i(x)|^2 + \sum_n \left| \sum_i y_i \frac{\partial f_i(x)}{\partial x_n} \right|^2. \quad (26.5.5)$$

The potential is always minimized by choosing the x_n to minimize the first term; whatever values this gives the x_n , the second term can always be minimized by taking $y_i = 0$. Whether or not supersymmetry can be spontaneously broken, these models have the peculiar feature that there are always directions in the space of the fields in which the minimum of the potential is flat. Whatever values x_{n0} of the x_n minimize the first term in Eq. (26.5.5), the second term vanishes not only for $y_i = 0$, but for any vector y_i in a direction orthogonal to all the vectors $(v^n)_i = (\partial f_i(x)/\partial x_n)_{x=x_0}$. If there are N_X superfields X_n and N_Y superfields Y_i with $N_Y > N_X$, then the v^n cannot span the space of the y s, and there will be at least $N_Y - N_X$ of these flat directions. For any non-vanishing values $y_i = y_{0i}$ along any one of these flat directions, the R -symmetry of the Lagrangian density is spontaneously broken, and the Goldstone boson field ϕ associated with this global symmetry breakdown corresponds to a term $\propto y_{0i}$ in the y_i .

The simplest example of this class of models is provided by the case where there is just one X superfield and two Y superfields. Renormalizability requires the coefficient functions $f_i(X)$ to be quadratic functions of X , and by taking suitable linear combinations of the Y_i and shifting and rescaling X we can choose these functions so that

$$f_1(X) = X - a, \quad f_2(X) = X^2, \quad (26.5.6)$$

with an arbitrary constant a . There clearly is no simultaneous solution of the two equations (26.5.1) unless the superpotential is fine-tuned so that $a = 0$. The potential (26.5.5) here is

$$V(x, y) = |x|^4 + |x - a|^2 + |y_1 + 2xy_2|^2. \quad (26.5.7)$$

The sum of the first two terms has a unique global minimum x_0 . The flat direction here is the one for which $y_1 + 2x_0y_2 = 0$. For $a = 0$ we have $x_0 = 0$, and the minima of the potential are along the line with $y_1 = 0$ and y_2 arbitrary.

Whatever the reason for a spontaneous breakdown of supersymmetry, this phenomenon always entails the existence of a massless spin 1/2 particle, the *goldstino*, analogous to the Goldstone bosons associated with the spontaneous breakdown of ordinary global symmetries. (The one exception, discussed in Section 31.3, is that in supergravity theories, where supersymmetry is a local symmetry, the goldstino appears as the helicity $\pm 1/2$ states of a massive particle of spin 3/2, the gravitino.) In renormalizable theories of chiral superfields, the tree-approximation vacuum expectation values ϕ_{n0} of the scalar fields must be at a minimum of the potential $\sum_n |\partial f(\phi)/\partial \phi_n|^2$ in Eq. (26.4.7), so

$$\sum_m \mathcal{M}_{nm} \left(\frac{\partial f(\phi)}{\partial \phi_m} \Big|_{\phi=\phi_0} \right)^* = 0, \quad (26.5.8)$$

where

$$\mathcal{M}_{nm} \equiv \frac{\partial^2 f(\phi)}{\partial \phi_n \partial \phi_m} \Big|_{\phi=\phi_0}. \quad (26.5.9)$$

If Eq. (26.5.1) is not satisfied, then Eq. (26.5.8) tells us that the matrix $\mathcal{M}_{nm}$ has at least one eigenvector with eigenvalue zero, so according to Eq. (26.4.10) there must be at least one linear combination of the spin 1/2 particles described by ψ_n with zero mass. For instance,

for the model defined by Eqs. (26.5.2) and (26.5.6), the matrix $\mathcal{M}$ has the non-vanishing components

$$\mathcal{M}_{xy_1} = \mathcal{M}_{y_1x} = 1, \quad \mathcal{M}_{xy_2} = \mathcal{M}_{y_2x} = 2x_0, \quad (26.5.10)$$

so this matrix has eigenvalues $\pm 2x_0$ and 0, with the last eigenvalue corresponding to the goldstino mode. In Chapter 29 we will show without the use of perturbation theory that the spontaneous breakdown of supersymmetry requires the existence of goldstinos, and explore their general properties.